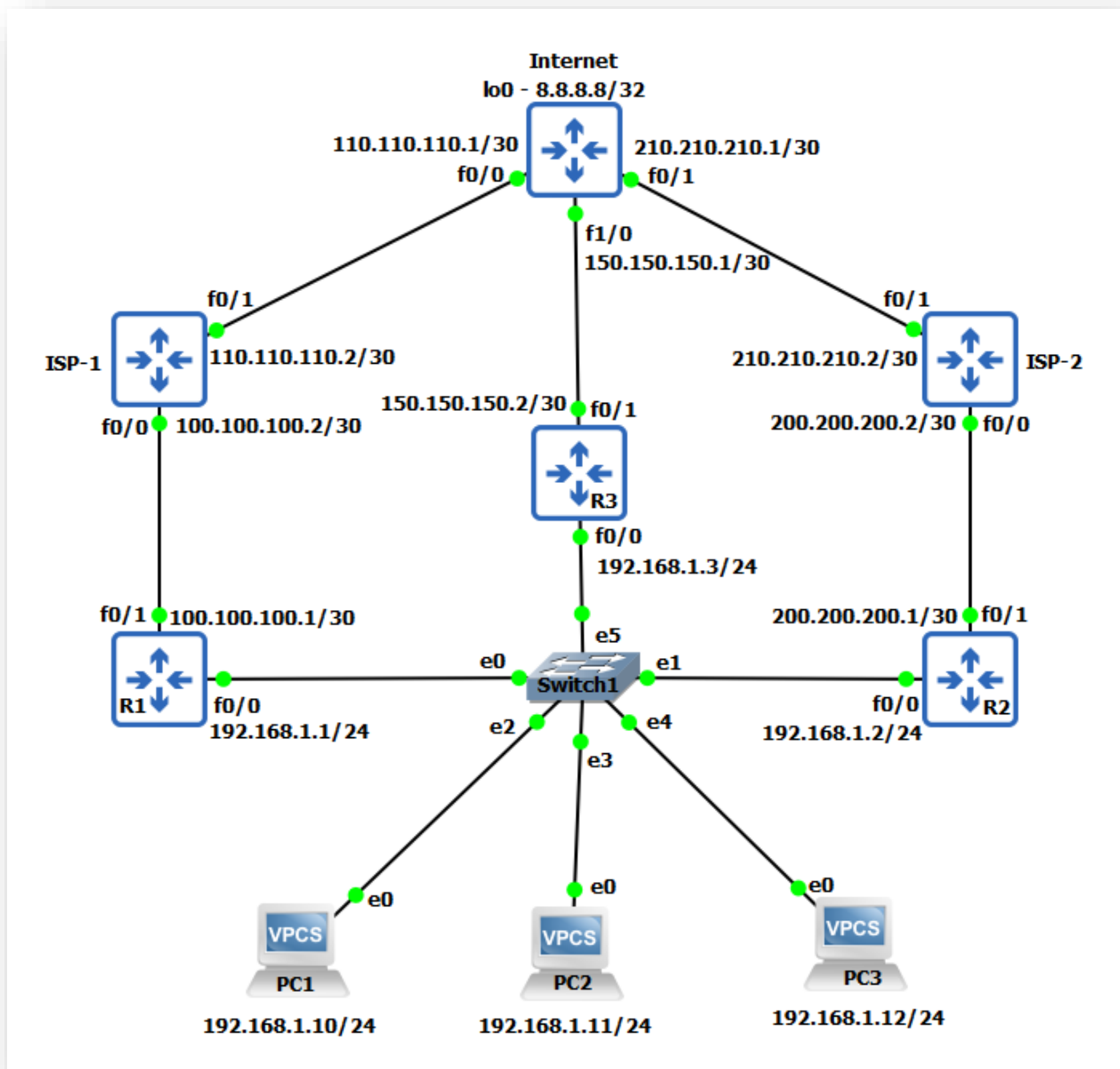


Gateway Load Balance Protocol (GLBP)



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna**. (Refer Lab-22 for task-1 to task-5)
2. Configure VPCs with IP address and default gateway of 192.168.1.4 (GLBP Virtual IP)
3. Configure default route on R1, R2 & R3 pointing to ISP
4. Configure OSPF on ISP-1, ISP-2 and Internet router.
5. Configure NAT overload (PAT) on R1, R2 and R3
6. Configure GLBP with virtual IP 192.168.1.4 such that R3 will be Active, R2 will be Standby and R3 will be in Listen mode.
7. Configure Weight such that R3 will handle twice the requests than R1 & R2.

Task-6: Configure GLBP with virtual IP 192.168.1.4 such that R3 will be Active, R2 will be Standby and R3 will be in Listen mode.

R1 Configuration:

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#glbp 1 ip 192.168.1.4
R1(config-if)#glbp 1 name GLBP
R1(config-if)#glbp 1 priority 105
R1(config-if)#glbp 1 preempt
R1(config-if)#exit
R1(config)#exit
R1#
```

R2 Configuration:

```
R2#config t
R2(config)#int fa0/0
R2(config-if)#glbp 1 ip 192.168.1.4
R2(config-if)#glbp 1 name GLBP
R2(config-if)#glbp 1 preempt
R2(config-if)#exit
R2(config)#exit
R2#
```

R3 Configuration:

```
R3#config t
R3(config)#int fa0/0
R3(config-if)#glbp 1 ip 192.168.1.4
R3(config-if)#glbp 1 name GLBP
R3(config-if)#glbp 1 priority 110
R3(config-if)#glbp 1 preempt
R3(config-if)#exit
R3(config)#exit
R3#
```

✓ Verification & Testing:

R1#sh glbp

FastEthernet0/0 - Group 1

State is Standby

3 state changes, last state change 00:01:28

Virtual IP address is 192.168.1.4

Hello time 3 sec, hold time 10 sec

Next hello sent in 1.312 secs

Redirect time 600 sec, forwarder timeout 14400 sec

Preemption enabled, min delay 0 sec

Active is 192.168.1.3, priority 110 (expires in 10.144 sec)

Standby is local

Priority 105 (configured)

Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1) local

ca02.02b4.0008 (192.168.1.2)

ca06.3008.0008 (192.168.1.3)

There are 3 forwarders (1 active)

Forwarder 1

State is Active

1 state change, last state change 00:03:35

MAC address is 0007.b400.0101 (default)

Owner ID is ca01.25e0.0008

Preemption enabled, min delay 30 sec

Active is local, weighting 100

Forwarder 2

State is Listen

MAC address is 0007.b400.0102 (learnt)

Owner ID is ca02.02b4.0008

Time to live: 14397.472 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.2 (primary), weighting 100 (expires in 8.064 sec)

Forwarder 3

State is Listen

2 state changes, last state change 00:00:55

MAC address is 0007.b400.0103 (learnt)

Owner ID is ca06.3008.0008

Time to live: 14398.400 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.3 (primary), weighting 100 (expires in 10.336 sec)

R1#

R2#sh glbp

FastEthernet0/0 - Group 1

State is Listen

2 state changes, last state change 00:02:22

Virtual IP address is 192.168.1.4

Hello time 3 sec, hold time 10 sec

Next hello sent in 1.536 secs

Redirect time 600 sec, forwarder timeout 14400 sec

Preemption enabled, min delay 0 sec

Active is 192.168.1.3, priority 110 (expires in 11.520 sec)

Standby is 192.168.1.1, priority 105 (expires in 8.064 sec)

Priority 100 (default)

Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1)

ca02.02b4.0008 (192.168.1.2) local

ca06.3008.0008 (192.168.1.3)

There are 3 forwarders (1 active)

Forwarder 1

State is Listen

MAC address is 0007.b400.0101 (learnt)

Owner ID is ca01.25e0.0008

Time to live: 14398.080 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.1 (primary), weighting 100 (expires in 8.704 sec)

Forwarder 2

State is Active

1 state change, last state change 00:02:54

MAC address is 0007.b400.0102 (default)

Owner ID is ca02.02b4.0008

Preemption enabled, min delay 30 sec

Active is local, weighting 100

Forwarder 3

State is Listen

MAC address is 0007.b400.0103 (learnt)

Owner ID is ca06.3008.0008

Time to live: 14399.936 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.3 (primary), weighting 100 (expires in 10.976 sec)

R2#

R3#sh glbp

FastEthernet0/0 - Group 1

State is Active

1 state change, last state change 00:03:09

Virtual IP address is 192.168.1.4

Hello time 3 sec, hold time 10 sec

Next hello sent in 2.464 secs

Redirect time 600 sec, forwarder timeout 14400 sec

Preemption enabled, min delay 0 sec

Active is local

Standby is 192.168.1.1, priority 105 (expires in 8.096 sec)

Priority 110 (configured)

Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1)

ca02.02b4.0008 (192.168.1.2)

ca06.3008.0008 (192.168.1.3) local

There are 3 forwarders (1 active)

Forwarder 1

State is Listen

MAC address is 0007.b400.0101 (learnt)

Owner ID is ca01.25e0.0008

Redirection enabled, 598.112 sec remaining (maximum 600 sec)

Time to live: 14398.112 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.1 (primary), weighting 100 (expires in 9.408 sec)

Forwarder 2

State is Listen

MAC address is 0007.b400.0102 (learnt)

Owner ID is ca02.02b4.0008

Redirection enabled, 599.360 sec remaining (maximum 600 sec)

Time to live: 14399.360 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.2 (primary), weighting 100 (expires in 11.072 sec)

Forwarder 3

State is Active

1 state change, last state change 00:02:26

MAC address is 0007.b400.0103 (default)

Owner ID is ca06.3008.0008

Redirection enabled

Preemption enabled, min delay 30 sec

Active is local, weighting 100

R3#

R1#sh glbp brief

Interface	Grp	Fwd	Pri	State	Address	Active router	Standby router
Fa0/0	1	-	105	Standby	192.168.1.4	192.168.1.3	local
Fa0/0	1	1	-	Active	0007.b400.0101	local	-
Fa0/0	1	2	-	Listen	0007.b400.0102	192.168.1.2	-
Fa0/0	1	3	-	Listen	0007.b400.0103	192.168.1.3	-

R1#

R2#sh glbp brief

Interface	Grp	Fwd	Pri	State	Address	Active router	Standby router
Fa0/0	1	-	100	Listen	192.168.1.4	192.168.1.3	192.168.1.1
Fa0/0	1	1	-	Listen	0007.b400.0101	192.168.1.1	-
Fa0/0	1	2	-	Active	0007.b400.0102	local	-
Fa0/0	1	3	-	Listen	0007.b400.0103	192.168.1.3	-

R2#

R3#sh glbp brief

Interface	Grp	Fwd	Pri	State	Address	Active router	Standby router
Fa0/0	1	-	110	Active	192.168.1.4	local	192.168.1.1
Fa0/0	1	1	-	Listen	0007.b400.0101	192.168.1.1	-
Fa0/0	1	2	-	Listen	0007.b400.0102	192.168.1.2	-
Fa0/0	1	3	-	Active	0007.b400.0103	local	-

R3#

Ping GLBP Virtual IP 192.168.1.4 from PCs

PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=255 time=10.442 ms

84 bytes from 192.168.1.4 icmp_seq=2 ttl=255 time=7.661 ms

84 bytes from 192.168.1.4 icmp_seq=3 ttl=255 time=6.406 ms

84 bytes from 192.168.1.4 icmp_seq=4 ttl=255 time=5.546 ms

84 bytes from 192.168.1.4 icmp_seq=5 ttl=255 time=2.398 ms

PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=255 time=8.396 ms
 84 bytes from 192.168.1.4 icmp_seq=2 ttl=255 time=2.276 ms
 84 bytes from 192.168.1.4 icmp_seq=3 ttl=255 time=10.431 ms
 84 bytes from 192.168.1.4 icmp_seq=4 ttl=255 time=12.550 ms
 84 bytes from 192.168.1.4 icmp_seq=5 ttl=255 time=10.466 ms

PC3> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=255 time=8.448 ms
 84 bytes from 192.168.1.4 icmp_seq=2 ttl=255 time=11.975 ms
 84 bytes from 192.168.1.4 icmp_seq=3 ttl=255 time=2.452 ms
 84 bytes from 192.168.1.4 icmp_seq=4 ttl=255 time=9.541 ms
 84 bytes from 192.168.1.4 icmp_seq=5 ttl=255 time=8.444 ms

Ping and trace Internet IP 8.8.8.8 from PCs**PC1> ping 8.8.8.8**

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=44.257 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=51.452 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=51.430 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=54.316 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=46.281 ms

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

1 192.168.1.1 3.451 ms 9.444 ms 9.388 ms
 2 100.100.100.2 32.377 ms 31.475 ms 32.286 ms
 3 *110.110.110.1 53.448 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> sh arp

00:07:b4:00:01:01 192.168.1.4 expires in 80 seconds

PC2> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=46.461 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=52.210 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=50.251 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=47.324 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=50.366 ms

PC2> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

1 192.168.1.2 9.284 ms 10.217 ms 9.203 ms
 2 200.200.200.2 32.249 ms 32.243 ms 31.221 ms
 3 *210.210.210.1 52.133 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> sh arp

00:07:b4:00:01:02 192.168.1.4 expires in 73 seconds

PC3> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=254 time=31.304 ms
 84 bytes from 8.8.8.8 icmp_seq=2 ttl=254 time=33.348 ms
 84 bytes from 8.8.8.8 icmp_seq=3 ttl=254 time=24.581 ms
 84 bytes from 8.8.8.8 icmp_seq=4 ttl=254 time=31.507 ms
 84 bytes from 8.8.8.8 icmp_seq=5 ttl=254 time=30.540 ms

PC3> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.3 11.525 ms 9.555 ms 10.531 ms
- 2 *150.150.150.1 31.429 ms (ICMP type:3, code:3, Destination port unreachable)

PC3> sh arp

00:07:b4:00:01:03 192.168.1.4 expires in 101 seconds

R1#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	100.100.100.1:38015	192.168.1.10:38015	8.8.8.8:38015	8.8.8.8:38015
icmp	100.100.100.1:38271	192.168.1.10:38271	8.8.8.8:38271	8.8.8.8:38271
icmp	100.100.100.1:38527	192.168.1.10:38527	8.8.8.8:38527	8.8.8.8:38527
icmp	100.100.100.1:38783	192.168.1.10:38783	8.8.8.8:38783	8.8.8.8:38783
icmp	100.100.100.1:39039	192.168.1.10:39039	8.8.8.8:39039	8.8.8.8:39039

R2#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	200.200.200.1:47743	192.168.1.11:47743	8.8.8.8:47743	8.8.8.8:47743
icmp	200.200.200.1:47999	192.168.1.11:47999	8.8.8.8:47999	8.8.8.8:47999
icmp	200.200.200.1:48255	192.168.1.11:48255	8.8.8.8:48255	8.8.8.8:48255
icmp	200.200.200.1:48511	192.168.1.11:48511	8.8.8.8:48511	8.8.8.8:48511
icmp	200.200.200.1:48767	192.168.1.11:48767	8.8.8.8:48767	8.8.8.8:48767

R3#sh ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	150.150.150.2:54399	192.168.1.12:54399	8.8.8.8:54399	8.8.8.8:54399
icmp	150.150.150.2:54655	192.168.1.12:54655	8.8.8.8:54655	8.8.8.8:54655
icmp	150.150.150.2:55167	192.168.1.12:55167	8.8.8.8:55167	8.8.8.8:55167
icmp	150.150.150.2:55423	192.168.1.12:55423	8.8.8.8:55423	8.8.8.8:55423
icmp	150.150.150.2:55679	192.168.1.12:55679	8.8.8.8:55679	8.8.8.8:55679

R1#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.100.100.1	-	ca01.25e0.0006	ARPA	FastEthernet0/1
Internet	100.100.100.2	82	ca03.09c8.0008	ARPA	FastEthernet0/1
Internet	192.168.1.1	-	ca01.25e0.0008	ARPA	FastEthernet0/0
Internet	192.168.1.2	47	ca02.02b4.0008	ARPA	FastEthernet0/0
Internet	192.168.1.4	-	0007.b400.0101	ARPA	FastEthernet0/0
Internet	192.168.1.10	0	0050.7966.6800	ARPA	FastEthernet0/0
Internet	192.168.1.11	0	0050.7966.6801	ARPA	FastEthernet0/0
Internet	192.168.1.12	0	0050.7966.6802	ARPA	FastEthernet0/0

R1#

R2#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.1.1	0	ca01.25e0.0008	ARPA	FastEthernet0/0
Internet	192.168.1.2	-	ca02.02b4.0008	ARPA	FastEthernet0/0
Internet	192.168.1.4	-	0007.b400.0102	ARPA	FastEthernet0/0
Internet	192.168.1.10	0	0050.7966.6800	ARPA	FastEthernet0/0
Internet	192.168.1.11	0	0050.7966.6801	ARPA	FastEthernet0/0
Internet	200.200.200.1	-	ca02.02b4.0006	ARPA	FastEthernet0/1
Internet	200.200.200.2	82	ca04.1438.0008	ARPA	FastEthernet0/1

R2#

R3#sh arp

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	150.150.150.1	44	ca05.18a0.001c	ARPA	FastEthernet0/1
Internet	150.150.150.2	-	ca06.3008.0006	ARPA	FastEthernet0/1
Internet	192.168.1.3	-	ca06.3008.0008	ARPA	FastEthernet0/0
Internet	192.168.1.4	-	0007.b400.0103	ARPA	FastEthernet0/0
Internet	192.168.1.10	0	0050.7966.6800	ARPA	FastEthernet0/0
Internet	192.168.1.11	0	0050.7966.6801	ARPA	FastEthernet0/0
Internet	192.168.1.12	0	0050.7966.6802	ARPA	FastEthernet0/0

R3#

Test – From all PC start continuous PING for 8.8.8.8 and simulate link failure by shutting down the R3's LAN interface.

```
R3#config t
R3(config)#int fa0/0
R3(config-if)#shut
R3(config-if)#exit
R3(config)#exit
R3#
```

✓ Verification & Testing:

PC1> ping 8.8.8.8 -t

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=52.250 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=52.474 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=47.412 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=51.282 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=49.579 ms
84 bytes from 8.8.8.8 icmp_seq=6 ttl=253 time=45.106 ms
84 bytes from 8.8.8.8 icmp_seq=7 ttl=253 time=46.477 ms
84 bytes from 8.8.8.8 icmp_seq=8 ttl=253 time=46.427 ms
84 bytes from 8.8.8.8 icmp_seq=9 ttl=253 time=45.428 ms
84 bytes from 8.8.8.8 icmp_seq=10 ttl=253 time=44.449 ms
84 bytes from 8.8.8.8 icmp_seq=11 ttl=253 time=45.333 ms
84 bytes from 8.8.8.8 icmp_seq=12 ttl=253 time=55.104 ms
84 bytes from 8.8.8.8 icmp_seq=13 ttl=253 time=48.288 ms
84 bytes from 8.8.8.8 icmp_seq=14 ttl=253 time=51.241 ms
84 bytes from 8.8.8.8 icmp_seq=15 ttl=253 time=54.373 ms
84 bytes from 8.8.8.8 icmp_seq=16 ttl=253 time=46.420 ms
```


84 bytes from 8.8.8.8 icmp_seq=17 ttl=253 time=49.459 ms
84 bytes from 8.8.8.8 icmp_seq=18 ttl=253 time=54.441 ms
84 bytes from 8.8.8.8 icmp_seq=19 ttl=253 time=53.358 ms
84 bytes from 8.8.8.8 icmp_seq=20 ttl=253 time=48.420 ms
84 bytes from 8.8.8.8 icmp_seq=21 ttl=253 time=51.356 ms

PC2> ping 8.8.8.8 -t

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=53.346 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=51.241 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=48.390 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=47.584 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=55.157 ms
84 bytes from 8.8.8.8 icmp_seq=6 ttl=253 time=53.373 ms
84 bytes from 8.8.8.8 icmp_seq=7 ttl=253 time=53.309 ms
84 bytes from 8.8.8.8 icmp_seq=8 ttl=253 time=48.261 ms
84 bytes from 8.8.8.8 icmp_seq=9 ttl=253 time=53.256 ms
84 bytes from 8.8.8.8 icmp_seq=10 ttl=253 time=47.381 ms
84 bytes from 8.8.8.8 icmp_seq=11 ttl=253 time=48.262 ms
84 bytes from 8.8.8.8 icmp_seq=12 ttl=253 time=50.242 ms
84 bytes from 8.8.8.8 icmp_seq=13 ttl=253 time=51.088 ms
84 bytes from 8.8.8.8 icmp_seq=14 ttl=253 time=55.199 ms
84 bytes from 8.8.8.8 icmp_seq=15 ttl=253 time=51.421 ms

PC3> ping 8.8.8.8 -t

84 bytes from 8.8.8.8 icmp_seq=1 ttl=254 time=42.258 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=254 time=25.484 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=254 time=30.311 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=254 time=27.338 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=254 time=25.526 ms
8.8.8.8 icmp_seq=6 timeout
8.8.8.8 icmp_seq=7 timeout
8.8.8.8 icmp_seq=8 timeout
8.8.8.8 icmp_seq=9 timeout
84 bytes from 8.8.8.8 icmp_seq=10 ttl=253 time=60.234 ms
84 bytes from 8.8.8.8 icmp_seq=11 ttl=253 time=44.297 ms
84 bytes from 8.8.8.8 icmp_seq=12 ttl=253 time=47.393 ms
84 bytes from 8.8.8.8 icmp_seq=13 ttl=253 time=51.267 ms
84 bytes from 8.8.8.8 icmp_seq=14 ttl=253 time=49.286 ms
84 bytes from 8.8.8.8 icmp_seq=15 ttl=253 time=44.316 ms
84 bytes from 8.8.8.8 icmp_seq=16 ttl=253 time=50.321 ms

R1#sh glbp

FastEthernet0/0 - Group 1

State is Active

4 state changes, last state change 00:03:32

Virtual IP address is 192.168.1.4

Hello time 3 sec, hold time 10 sec

Next hello sent in 0.928 secs

Redirect time 600 sec, forwarder timeout 14400 sec

Preemption enabled, min delay 0 sec

Active is local

Standby is 192.168.1.2, priority 100 (expires in 8.320 sec)

Priority 105 (configured)

Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1) local

ca02.02b4.0008 (192.168.1.2)

There are 3 forwarders (1 active)

Forwarder 1

State is Active

1 state change, last state change 01:25:58

MAC address is 0007.b400.0101 (default)

Owner ID is ca01.25e0.0008

Redirection enabled

Preemption enabled, min delay 30 sec

Active is local, weighting 100

Forwarder 2

State is Listen

MAC address is 0007.b400.0102 (learnt)

Owner ID is ca02.02b4.0008

Redirection enabled, 598.336 sec remaining (maximum 600 sec)

Time to live: 14398.336 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.2 (primary), weighting 100 (expires in 8.704 sec)

Forwarder 3

State is Listen

4 state changes, last state change 00:03:32

MAC address is 0007.b400.0103 (learnt)

Owner ID is ca06.3008.0008

Redirection enabled, 376.320 sec remaining (maximum 378 sec)

Time to live: 14176.320 sec (maximum 14178 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.2 (secondary), weighting 100 (expires in 8.608 sec)

R1#

R2#sh glbp

FastEthernet0/0 - Group 1

State is Standby

3 state changes, last state change 00:08:54

Virtual IP address is 192.168.1.4

Hello time 3 sec, hold time 10 sec

Next hello sent in 0.672 secs

Redirect time 600 sec, forwarder timeout 14400 sec

Preemption enabled, min delay 0 sec

Active is 192.168.1.1, priority 105 (expires in 11.392 sec)

Standby is local

Priority 100 (default)

Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1)

ca02.02b4.0008 (192.168.1.2) local

There are 3 forwarders (2 active)

Forwarder 1

State is Listen

MAC address is 0007.b400.0101 (learnt)

Owner ID is ca01.25e0.0008

Time to live: 14399.904 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.1 (primary), weighting 100 (expires in 10.208 sec)

Forwarder 2

State is Active

1 state change, last state change 01:30:08

MAC address is 0007.b400.0102 (default)

Owner ID is ca02.02b4.0008

Preemption enabled, min delay 30 sec

Active is local, weighting 100

Forwarder 3

State is Active

1 state change, last state change 00:09:05

MAC address is 0007.b400.0103 (learnt)

Owner ID is ca06.3008.0008

Time to live: 13843.904 sec (maximum 14389 sec)

Preemption enabled, min delay 30 sec

Active is local, weighting 100

R2#

R3#sh glbp

FastEthernet0/0 - Group 1

State is Init (interface down)

2 state changes, last state change 00:11:36

Virtual IP address is 192.168.1.4

Hello time 3 sec, hold time 10 sec

Redirect time 600 sec, forwarder timeout 14400 sec

Preemption enabled, min delay 0 sec

Active is unknown

Standby is unknown

Priority 110 (configured)

Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca06.3008.0008 (192.168.1.3) local

There are 3 forwarders (0 active)

Forwarder 1

State is Init

MAC address is 0007.b400.0101 (learnt)

Owner ID is ca01.25e0.0008

Time to live: 13701.760 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is unknown

Client selection count: 7

Forwarder 2

State is Init

MAC address is 0007.b400.0102 (learnt)

Owner ID is ca02.02b4.0008

Time to live: 13702.880 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is unknown

Client selection count: 7

Forwarder 3

State is Init

2 state changes, last state change 00:11:36

MAC address is 0007.b400.0103 (default)

Owner ID is ca06.3008.0008

Preemption enabled, min delay 30 sec

Active is unknown

Client selection count: 7

R3#

R1#sh glbp brief

Interface	Grp	Fwd	Pri	State	Address	Active router	Standby router
Fa0/0	1	-	105	Active	192.168.1.4	local	192.168.1.2
Fa0/0	1	1	-	Active	0007.b400.0101	local	-
Fa0/0	1	2	-	Listen	0007.b400.0102	192.168.1.2	-
Fa0/0	1	3	-	Listen	0007.b400.0103	192.168.1.2	-

R1#

R2#sh glbp brief

Interface	Grp	Fwd	Pri	State	Address	Active router	Standby router
Fa0/0	1	-	100	Standby	192.168.1.4	192.168.1.1	local
Fa0/0	1	1	-	Listen	0007.b400.0101	192.168.1.1	-
Fa0/0	1	2	-	Active	0007.b400.0102	local	-
Fa0/0	1	3	-	Active	0007.b400.0103	local	-

R2#

R3#sh glbp brief

Interface	Grp	Fwd	Pri	State	Address	Active router	Standby router
Fa0/0	1	-	110	Init	192.168.1.4	unknown	unknown
Fa0/0	1	1	-	Init	0007.b400.0101	unknown	-
Fa0/0	1	2	-	Init	0007.b400.0102	unknown	-
Fa0/0	1	3	-	Init	0007.b400.0103	unknown	-

R3#

Ping and trace Internet IP 8.8.8.8 from PCs

PC1> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=54.188 ms

84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=48.196 ms

84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=52.367 ms

84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=55.224 ms

84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=54.152 ms

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.1 31.297 ms 9.610 ms 10.540 ms
- 2 100.100.100.2 30.320 ms 31.413 ms 32.292 ms
- 3 *110.110.110.1 51.164 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> sh arp

00:07:b4:00:01:01 192.168.1.4 expires in 96 seconds

PC2> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=61.340 ms

84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=51.364 ms

84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=51.306 ms

84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=53.374 ms

84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=50.387 ms

PC2> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.2 9.320 ms 9.337 ms 10.509 ms
- 2 200.200.200.2 41.351 ms 30.419 ms 31.307 ms
- 3 *210.210.210.1 52.251 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> sh arp

00:07:b4:00:01:02 192.168.1.4 expires in 87 seconds

PC3> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=253 time=51.294 ms

84 bytes from 8.8.8.8 icmp_seq=2 ttl=253 time=46.348 ms

84 bytes from 8.8.8.8 icmp_seq=3 ttl=253 time=49.321 ms

84 bytes from 8.8.8.8 icmp_seq=4 ttl=253 time=47.475 ms

84 bytes from 8.8.8.8 icmp_seq=5 ttl=253 time=48.329 ms

PC3> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.1 12.455 ms 10.609 ms 10.603 ms
- 2 100.100.100.2 31.345 ms 30.467 ms 31.341 ms
- 3 *110.110.110.1 53.322 ms (ICMP type:3, code:3, Destination port unreachable)

PC3> sh arp

00:07:b4:00:01:01 192.168.1.4 expires in 73 seconds

Test – Simulate link recovery by making R3's LAN interface up.

R3#config t

R3(config)#int fa0/0

R3(config-if)#no shut

```
R3(config-if)#exit
R3(config)#exit
R3#
```

✓ Verification & Testing:

R3#sh glbp

FastEthernet0/0 - Group 1

State is Active

3 state changes, last state change 00:02:24
 Virtual IP address is 192.168.1.4
 Hello time 3 sec, hold time 10 sec
 Next hello sent in 0.384 secs
 Redirect time 600 sec, forwarder timeout 14400 sec
 Preemption enabled, min delay 0 sec
 Active is local
 Standby is 192.168.1.1, priority 105 (expires in 8.416 sec)
 Priority 110 (configured)
 Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1)
 ca02.02b4.0008 (192.168.1.2)
 ca06.3008.0008 (192.168.1.3) local

There are 3 forwarders (1 active)

Forwarder 1

State is Listen
 MAC address is 0007.b400.0101 (learnt)
 Owner ID is ca01.25e0.0008
 Redirection enabled, 598.432 sec remaining (maximum 600 sec)
 Time to live: 14398.432 sec (maximum 14400 sec)
 Preemption enabled, min delay 30 sec
 Active is 192.168.1.1 (primary), weighting 100 (expires in 8.800 sec)
 Client selection count: 7

Forwarder 2

State is Listen
 MAC address is 0007.b400.0102 (learnt)
 Owner ID is ca02.02b4.0008
 Redirection enabled, 598.816 sec remaining (maximum 600 sec)
 Time to live: 14398.816 sec (maximum 14400 sec)
 Preemption enabled, min delay 30 sec
 Active is 192.168.1.2 (primary), weighting 100 (expires in 9.600 sec)
 Client selection count: 7

Forwarder 3

State is Active
 3 state changes, last state change 00:01:57
 MAC address is 0007.b400.0103 (default)
 Owner ID is ca06.3008.0008
 Redirection enabled
 Preemption enabled, min delay 30 sec

Active is local, weighting 100
Client selection count: 7

R3#

R2#sh glbp

FastEthernet0/0 - Group 1

State is Listen

4 state changes, last state change 00:02:50
Virtual IP address is 192.168.1.4
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.256 secs
Redirect time 600 sec, forwarder timeout 14400 sec
Preemption enabled, min delay 0 sec
Active is 192.168.1.3, priority 110 (expires in 9.728 sec)
Standby is 192.168.1.1, priority 105 (expires in 9.632 sec)
Priority 100 (default)
Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1)
ca02.02b4.0008 (192.168.1.2) local
ca06.3008.0008 (192.168.1.3)

There are 3 forwarders (1 active)

Forwarder 1

State is Listen
MAC address is 0007.b400.0101 (learnt)
Owner ID is ca01.25e0.0008
Time to live: 14399.648 sec (maximum 14400 sec)
Preemption enabled, min delay 30 sec
Active is 192.168.1.1 (primary), weighting 100 (expires in 10.144 sec)

Forwarder 2

State is Active
1 state change, last state change 02:09:27
MAC address is 0007.b400.0102 (default)
Owner ID is ca02.02b4.0008
Preemption enabled, min delay 30 sec
Active is local, weighting 100

Forwarder 3

State is Listen
2 state changes, last state change 00:02:20
MAC address is 0007.b400.0103 (learnt)
Owner ID is ca06.3008.0008
Time to live: 14398.208 sec (maximum 14400 sec)
Preemption enabled, min delay 30 sec
Active is 192.168.1.3 (primary), weighting 100 (expires in 8.416 sec)

R2#

R1#sh glbp

FastEthernet0/0 - Group 1

State is Standby

6 state changes, last state change 00:03:01
 Virtual IP address is 192.168.1.4
 Hello time 3 sec, hold time 10 sec
 Next hello sent in 1.632 secs
 Redirect time 600 sec, forwarder timeout 14400 sec
 Preemption enabled, min delay 0 sec
 Active is 192.168.1.3, priority 110 (expires in 10.240 sec)
 Standby is local
 Priority 105 (configured)
 Weighting 100 (default 100), thresholds: lower 1, upper 100

Load balancing: round-robin

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1) local
 ca02.02b4.0008 (192.168.1.2)
 ca06.3008.0008 (192.168.1.3)

There are 3 forwarders (1 active)

Forwarder 1

State is Active

1 state change, last state change 02:11:13

MAC address is 0007.b400.0101 (default)

Owner ID is ca01.25e0.0008

Preemption enabled, min delay 30 sec

Active is local, weighting 100

Client selection count: 2

Forwarder 2

State is Listen

MAC address is 0007.b400.0102 (learnt)

Owner ID is ca02.02b4.0008

Time to live: 14399.008 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.2 (primary), weighting 100 (expires in 10.656 sec)

Client selection count: 1

Forwarder 3

State is Listen

4 state changes, last state change 00:48:46

MAC address is 0007.b400.0103 (learnt)

Owner ID is ca06.3008.0008

Time to live: 14399.808 sec (maximum 14400 sec)

Preemption enabled, min delay 30 sec

Active is 192.168.1.3 (primary), weighting 100 (expires in 10.496 sec)

R1#

Task-7: Configure Weight such that R3 will handle twice the requests than R1 & R2.

R1 Configuration:

R1#config t

R1(config)#int fa0/0

R1(config-if)#glbp 1 load-balancing weighted

```
R1(config-if)#glbp 1 weight 1
R1(config-if)#exit
R1(config)#exit
R1#
```

R2 Configuration:

```
R2#config t
R2(config)#int fa0/0
R2(config-if)#glbp 1 load-balancing weighted
R2(config-if)#glbp 1 weight 1
R2(config-if)#exit
R2(config)#exit
R2#
```

R3 Configuration:

```
R3#config t
R3(config)#int fa0/0
R3(config-if)#glbp 1 load-balancing weighted
R3(config-if)#glbp 1 weight 2
R3(config-if)#exit
R3(config)#exit
R3#
```

✓ Verification & Testing:**R1#sh glbp**

FastEthernet0/0 - Group 1

State is Standby

6 state changes, last state change 01:53:49
Virtual IP address is 192.168.1.4
Hello time 3 sec, hold time 10 sec
Next hello sent in 0.224 secs
Redirect time 600 sec, forwarder timeout 14400 sec
Preemption enabled, min delay 0 sec
Active is 192.168.1.3, priority 110 (expires in 9.856 sec)
Standby is local
Priority 105 (configured)
Weighting 1 (configured 1), thresholds: lower 1, upper 1
Load balancing: weighted
IP redundancy name is "GLBP"
Group members:
ca01.25e0.0008 (192.168.1.1) local
ca02.02b4.0008 (192.168.1.2)
ca06.3008.0008 (192.168.1.3)
There are 3 forwarders (1 active)
Forwarder 1
State is Active
1 state change, last state change 04:02:01
MAC address is 0007.b400.0101 (default)

Owner ID is ca01.25e0.0008
 Preemption enabled, min delay 30 sec
Active is local, weighting 1
 Client selection count: 2
 Forwarder 2
 State is Listen
 MAC address is 0007.b400.0102 (learnt)
 Owner ID is ca02.02b4.0008
 Time to live: 14399.616 sec (maximum 14400 sec)
 Preemption enabled, min delay 30 sec
Active is 192.168.1.2 (primary), weighting 1 (expires in 11.424 sec)
 Client selection count: 1
 Forwarder 3
 State is Listen
 4 state changes, last state change 02:39:35
 MAC address is 0007.b400.0103 (learnt)
 Owner ID is ca06.3008.0008
 Time to live: 14398.144 sec (maximum 14400 sec)
 Preemption enabled, min delay 30 sec
Active is 192.168.1.3 (primary), weighting 2 (expires in 9.248 sec)

R1#

R2#sh glbp

FastEthernet0/0 - Group 1

State is Listen
 4 state changes, last state change 02:00:15
 Virtual IP address is 192.168.1.4
 Hello time 3 sec, hold time 10 sec
 Next hello sent in 1.600 secs
 Redirect time 600 sec, forwarder timeout 14400 sec
 Preemption enabled, min delay 0 sec
 Active is 192.168.1.3, priority 110 (expires in 7.840 sec)
 Standby is 192.168.1.1, priority 105 (expires in 8.320 sec)
 Priority 100 (default)
Weighting 1 (configured 1), thresholds: lower 1, upper 1
Load balancing: weighted

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1)
 ca02.02b4.0008 (192.168.1.2) local
 ca06.3008.0008 (192.168.1.3)

There are 3 forwarders (1 active)

Forwarder 1

State is Listen
 MAC address is 0007.b400.0101 (learnt)
 Owner ID is ca01.25e0.0008
 Time to live: 14398.336 sec (maximum 14400 sec)
 Preemption enabled, min delay 30 sec

Active is 192.168.1.1 (primary), weighting 1 (expires in 8.416 sec)

Forwarder 2

State is Active

1 state change, last state change 04:06:51
MAC address is 0007.b400.0102 (default)
Owner ID is ca02.02b4.0008
Preemption enabled, min delay 30 sec

Active is local, weighting 1

Forwarder 3

State is Listen

2 state changes, last state change 01:59:45
MAC address is 0007.b400.0103 (learnt)
Owner ID is ca06.3008.0008
Time to live: 14397.504 sec (maximum 14400 sec)
Preemption enabled, min delay 30 sec

Active is 192.168.1.3 (primary), weighting 2 (expires in 8.480 sec)

R2#

R3#sh glbp

FastEthernet0/0 - Group 1

State is Active

3 state changes, last state change 02:18:23
Virtual IP address is 192.168.1.4
Hello time 3 sec, hold time 10 sec
Next hello sent in 1.376 secs
Redirect time 600 sec, forwarder timeout 14400 sec
Preemption enabled, min delay 0 sec
Active is local
Standby is 192.168.1.1, priority 105 (expires in 8.736 sec)
Priority 110 (configured)

Weighting 2 (configured 2), thresholds: lower 1, upper 2

Load balancing: weighted

IP redundancy name is "GLBP"

Group members:

ca01.25e0.0008 (192.168.1.1)
ca02.02b4.0008 (192.168.1.2)
ca06.3008.0008 (192.168.1.3) local

There are 3 forwarders (1 active)

Forwarder 1

State is Listen

MAC address is 0007.b400.0101 (learnt)
Owner ID is ca01.25e0.0008
Redirection enabled, 598.752 sec remaining (maximum 600 sec)
Time to live: 14398.752 sec (maximum 14400 sec)
Preemption enabled, min delay 30 sec

Active is 192.168.1.1 (primary), weighting 1 (expires in 9.856 sec)

Client selection count: 7

Forwarder 2

State is Listen

MAC address is 0007.b400.0102 (learnt)
Owner ID is ca02.02b4.0008
Redirection enabled, 598.624 sec remaining (maximum 600 sec)
Time to live: 14398.624 sec (maximum 14400 sec)
Preemption enabled, min delay 30 sec

Active is 192.168.1.2 (primary), weighting 1 (expires in 9.184 sec)

Client selection count: 7

Forwarder 3

State is Active

3 state changes, last state change 02:17:55

MAC address is 0007.b400.0103 (default)

Owner ID is ca06.3008.0008

Redirection enabled

Preemption enabled, min delay 30 sec

Active is local, weighting 2

Client selection count: 7

R3#

✓ Verification & Testing:

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.1 10.354 ms 9.225 ms 9.389 ms
- 2 100.100.100.2 30.425 ms 31.209 ms 32.136 ms
- 3 *110.110.110.1 53.176 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> sh arp

00:07:b4:00:01:01 192.168.1.4 expires in 86 seconds

PC1> clear arp

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.2 9.282 ms 10.306 ms 9.341 ms
- 2 200.200.200.2 32.259 ms 31.326 ms 32.316 ms
- 3 *210.210.210.1 51.287 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> sh arp

00:07:b4:00:01:02 192.168.1.4 expires in 113 seconds

PC1> clear arp

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.3 20.380 ms 8.310 ms 10.372 ms
- 2 *150.150.150.1 31.260 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> sh arp

00:07:b4:00:01:03 192.168.1.4 expires in 116 seconds

PC1> clear arp

PC1> trace 8.8.8.8

trace to 8.8.8.8, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.3 9.265 ms 10.338 ms 10.502 ms
- 2 *150.150.150.1 30.406 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> sh arp

00:07:b4:00:01:03 192.168.1.4 expires in 110 seconds

To Remove GLBP

```
R1#config t
R1(config)#int fa0/0
R1(config-if)#no glbp 1 ip 192.168.1.4
R1(config-if)#no glbp 1 name GLBP
R1(config-if)#no glbp 1 priority 105
R1(config-if)#no glbp 1 preempt
R1(config-if)#no glbp 1 load-balancing weighted
R1(config-if)#no glbp 1 weight 1
R1(config-if)#exit
R1(config)#exit
R1#
```

```
R2#config t
R2(config)#int fa0/0
R2(config-if)#no glbp 1 ip 192.168.1.4
R2(config-if)#no glbp 1 name GLBP
R2(config-if)#no glbp 1 preempt
R2(config-if)#no glbp 1 load-balancing weighted
R2(config-if)#no glbp 1 weight 1
R2(config-if)#exit
R2(config)#exit
R2#
```

```
R3# config t
R3(config)#int fa0/0
R3(config-if)#no glbp 1 ip 192.168.1.4
R3(config-if)#no glbp 1 name GLBP
R3(config-if)#no glbp 1 priority 110
R3(config-if)#no glbp 1 preempt
R3(config-if)#no glbp 1 load-balancing weighted
R3(config-if)#no glbp 1 weight 2
R3(config-if)#exit
R3(config)#exit
R3#
```

Important Commands:

```
sh ip int brief | exclude unassign  
sh ip route static  
sh ip route ospf  
sh glbp  
sh glbp brief  
sh ip nat translations  
sh arp
```

Ratan